

Transactions in MySQL – Short Notes

What is a Transaction?

A **transaction** is a group of one or more SQL statements executed as a **single unit of work**.

It ensures that:

- **All operations succeed** → `COMMIT`
 - **If any operation fails, all changes are undone** → `ROLLBACK`
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Why Use Transactions?

- Maintain data consistency
 - Prevent partial updates
 - Recover from errors
 - Ensure data integrity
-

ACID Properties

1. Atomicity

A transaction is completed **entirely or not at all**.

2. Consistency

The database remains in a **valid state** before and after the transaction.

3. Isolation

Multiple transactions do not interfere with each other.

4. Durability

Once committed, changes are **permanent**, even if the system crashes.

Transaction Commands

1. START TRANSACTION

Begins a transaction.

```
START TRANSACTION;
```

or

```
BEGIN;
```

2. COMMIT

Saves all changes permanently.

```
COMMIT;
```

3. ROLLBACK

Undoes all uncommitted changes.

```
ROLLBACK;
```

4. SAVEPOINT

Creates a checkpoint within a transaction.

```
SAVEPOINT sp1;
```

Rollback to the savepoint:

```
ROLLBACK TO sp1;
```

5. RELEASE SAVEPOINT

Removes a savepoint.

```
RELEASE SAVEPOINT sp1;
```

Example

```
START TRANSACTION;
```

```
UPDATE accounts  
SET balance = balance - 500  
WHERE id = 1;
```

```
UPDATE accounts  
SET balance = balance + 500  
WHERE id = 2;
```

```
COMMIT;
```

If an error occurs:

```
ROLLBACK;
```

Autocommit

By default, MySQL automatically commits every SQL statement.

Check status:

```
SELECT @@autocommit;
```

Disable:

```
SET autocommit = 0;
```

Enable:

```
SET autocommit = 1;
```

Can Be Rolled Back

- INSERT
- UPDATE
- DELETE

Cannot Be Rolled Back (DDL)

- CREATE
- ALTER
- DROP
- TRUNCATE

These statements cause an **implicit COMMIT**.

Summary

Command	Purpose
START TRANSACTION / BEGIN	Start a transaction
COMMIT	Save changes permanently
ROLLBACK	Undo uncommitted changes

Command	Purpose
SAVEPOINT	Create a checkpoint
ROLLBACK TO	Undo changes after a savepoint
RELEASE SAVEPOINT	Delete a savepoint

Remember

- A **transaction** is a group of SQL statements treated as one unit.
- **COMMIT** saves changes permanently.
- **ROLLBACK** cancels uncommitted changes.
- **SAVEPOINT** allows partial rollback.
- MySQL's **InnoDB** storage engine supports transactions.